Linear Algebra

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1 Lessons

1.1 Linear maps: what they preserve, and what they forget

Lesson 1 described the static geometry of vector spaces. Now we study the maps between them. A system $A\mathbf{x} = \mathbf{b}$ asks two questions: does $\mathbf{b}$ lie among the achievable outputs of A, and if so, is $\mathbf{x}$ unique? Both questions are about the map $\mathbf{x} \mapsto A\mathbf{x}$. Differentiation is another map of the same shape: $D(p) = p'$ is linear, and it "forgets" exactly the constant polynomials. The kernel and the image make this precise.

Definition 1.1 (Linear map)**.** A function $T\colon V \to W$ between vector spaces is *linear* if

$$T(u+v) = T(u) + T(v), \qquad T(av) = aT(v)$$

for all $u, v \in V$ and $a \in \mathbb{R}$. Equivalently, $T(au+bv) = aT(u)+bT(v)$. By induction, T preserves every linear combination:

$$T(a_1v_1 + \cdots + a_kv_k) = a_1T(v_1) + \cdots + a_kT(v_k).$$

Taking $a = 0$ gives $T(0) = 0$: every linear map sends zero to zero.

In code terms, a linear map is a pure function that distributes over the two vector operations: the response to a combined input is the combination of the responses (superposition). The analogy breaks where programs branch: no thresholds, saturation, or case splits can be linear.

Example 1.2 (Linear maps and an impostor)**.** (a) Projection $P\colon \mathbb{R}^3 \to \mathbb{R}^3$, $P(x, y, z) = (x, y, 0)$, is linear. Its outputs all lie in the xy-plane, and it kills exactly the z-axis. We will revisit this in the language of kernels and images.

(b) Differentiation D on polynomials, $D(p) = p'$, is linear by the sum and constant-multiple rules for derivatives.

(c) The translation $S\colon \mathbb{R}^2 \to \mathbb{R}^2$, $S(x, y) = (x+1, y)$, is *not* linear: $S(0, 0) = (1, 0) \neq (0, 0)$. A translation is a linear map plus a fixed shift (an *affine* map). This is the same phenomenon as in Lesson 1: shifting a subspace off the origin destroys the structure.

Definition 1.3 (Kernel and image)**.** For a linear map $T\colon V \to W$, the *kernel* and *image* are

$$\ker T = \{v \in V : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

Note where they live: $\ker T \subseteq V$ (domain side), $\operatorname{im} T \subseteq W$ (codomain side).

Theorem 1.4 (Kernels and images are subspaces)**.** *If $T\colon V \to W$ is linear, then $\ker T$ is a subspace of V and $\operatorname{im} T$ is a subspace of W.*

Proof. Since $T(0) = 0$, we have $0 \in \ker T$, and $T(0) \in \operatorname{im} T$, so both sets are nonempty. If $T(u) = T(v) = 0$, then $T(au + bv) = aT(u) + bT(v) = 0$, so $\ker T$ passes the subspace test. If $x = T(u)$ and $y = T(v)$, then $ax + by = T(au + bv) \in \operatorname{im} T$, so $\operatorname{im} T$ passes it too. □

Example 1.5 (Kernel and image of a matrix map)**.** Recall the matrix from Lesson 1:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix}, \qquad T(\mathbf{x}) = A\mathbf{x}, \quad T\colon \mathbb{R}^3 \to \mathbb{R}^3.$$

Row reduction ($R_2 - 2R_1$, $R_3 - R_1$, $R_3 - R_2$, $R_1 - R_2$) gives pivots in columns 1 and 3, and $A\mathbf{x} = 0$ reduces to $x_1 + 2x_2 = 0$, $x_3 = 0$. Hence

$$\ker T = \operatorname{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \right\},$$

which has dimension 1. The image is the column space, spanned by the pivot columns of the *original* matrix (columns 1 and 3):

$$\operatorname{im} T = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \right\},$$

which has dimension 2. Note that $1+2 = 3$, the dimension of the domain. That is no coincidence.

Theorem 1.6 (A linear map is fixed by its values on a basis)**.** *Let $(v_1, \ldots, v_n)$ be a basis of V. For every choice of vectors $w_1, \ldots, w_n \in W$, there is exactly one linear map $T\colon V \to W$ with $T(v_i) = w_i$ for all i.*

Proof. Every $v \in V$ has unique coordinates $v = \sum_i a_i v_i$ (Lesson 1), so the formula $T(v) = \sum_i a_i w_i$ is well defined. It satisfies $T(v_i) = w_i$, and linearity follows directly from the coordinate arithmetic. Any linear S with $S(v_i) = w_i$ must satisfy $S(v) = \sum_i a_i w_i$ by preservation of combinations, so T is unique. □

Theorem 1.7 (Maps $\mathbb{R}^n \to \mathbb{R}^m$ are matrices)**.** *Every linear map $T\colon \mathbb{R}^n \to \mathbb{R}^m$ has the form $T(\mathbf{x}) = A\mathbf{x}$ for a unique $m \times n$ matrix A, whose j-th column is $T(e_j)$.*

Proof. The standard basis vectors $e_1, \ldots, e_n$ form a basis of $\mathbb{R}^n$, so by Theorem 1.6 the values $T(e_j)$ determine T. Writing $\mathbf{x} = \sum_j x_j e_j$ and using linearity,

$$T(\mathbf{x}) = \sum_j x_j T(e_j) = A\mathbf{x}, \qquad A = \begin{pmatrix} T(e_1) & \cdots & T(e_n) \end{pmatrix}.$$

□

A system $A\mathbf{x} = \mathbf{b}$ therefore asks: is $\mathbf{b}$ in $\operatorname{im} T$, and is the preimage unique?

Lemma 1.8 (Exchange lemma)**.** *If $(v_1, \dots, v_n)$ spans V and $(w_1, \dots, w_m)$ is linearly independent in V, then $m \le n$.*

Proof. Suppose $m > n$. We show by induction on $k = 0, \dots, n$ that, after reordering the v's, the list $w_1, \dots, w_k, v_{k+1}, \dots, v_n$ still spans V. This holds for $k = 0$. Assume it for some $k < n$ and write

$$w_{k+1} = a_1 w_1 + \cdots + a_k w_k + b_{k+1} v_{k+1} + \cdots + b_n v_n.$$

Not all b_j are zero: otherwise w_{k+1} would be a combination of $w_1, \dots, w_k$, contradicting independence. Reorder $v_{k+1}, \dots, v_n$ so that $b_{k+1} \neq 0$. Then v_{k+1} is a linear combination of $w_1, \dots, w_{k+1}, v_{k+2}, \dots, v_n$ (divide by b_{k+1}), so that list spans everything the old one did. At $k = n$, $w_1, \dots, w_n$ span V, so w_{n+1} is their linear combination, contradicting independence. Hence $m \le n$. □

Corollary 1.9 (Dimension is well defined)**.** *Any two finite bases of the same vector space have the same number of vectors. This closes the open claim from Lesson 1:* $\dim V$ *is unambiguous.*

Proof. Apply Theorem 1.8 twice: each basis spans, and the other is independent, so each size is $\le$ the other. □

Corollary 1.10 (Extending an independent list)**.** *In a space with a finite basis, every linearly independent list extends to a basis.*

Proof. Start with the independent list. Append the vectors of a finite basis one by one, keeping each new vector only when it lies outside the span of everything kept so far. The kept list stays independent by construction, and it spans V, since every basis vector is either kept or was already in the span when discarded. □

Definition 1.11 (Rank and nullity)**.** For a linear map $T\colon V \to W$, the *rank* is $\dim \operatorname{im} T$ and the *nullity* is $\dim \ker T$.

Theorem 1.12 (Rank–nullity)**.** *Let $T\colon V \to W$ be linear, with V having a finite basis of n vectors. Then*

$$\dim \ker T + \dim \operatorname{im} T = n.$$

Proof. Let $(u_1, \dots, u_k)$ be a basis of $\ker T$, and extend it by Theorem 1.10 to a basis $(u_1, \dots, u_n)$ of V. We claim that $T(u_{k+1}), \dots, T(u_n)$ is a basis of $\operatorname{im} T$. It spans: for $v = \sum_{i=1}^{n} a_i u_i$,

$$T(v) = \sum_{i=1}^{n} a_i T(u_i) = \sum_{i=k+1}^{n} a_i T(u_i),$$

since $T(u_i) = 0$ for $i \le k$. It is independent: if $\sum_{i=k+1}^{n} b_i T(u_i) = 0$, then $\sum_{i=k+1}^{n} b_i u_i \in \ker T$, so it equals $\sum_{j=1}^{k} c_j u_j$ for some coefficients c_j. Independence of the full basis $(u_1, \dots, u_n)$ forces every b_i (and every c_j) to be zero. Thus $\dim \operatorname{im} T = n - k$. □

Corollary 1.13 (Injectivity test)**.** *T is injective if and only if* $\ker T = \{0\}$.

Proof. $T(u) = T(v)$ holds if and only if $T(u - v) = 0$, so distinct inputs collide exactly when a nonzero vector lies in the kernel. □

For a matrix map $T(\mathbf{x}) = A\mathbf{x}$, rank–nullity is visible in row reduction: with n columns, each column is either a pivot column or a free column, so

$$(\#\text{ pivot columns}) + (\#\text{ free columns}) = n,$$

i.e. $\operatorname{rank} A + \dim \ker T = n$. The pivot *indices* come from the reduced form, but the basis of $\operatorname{im} T$ must be taken from those columns of the *original* matrix. Row operations preserve the kernel but generally change the column space, so reading an image basis off the reduced matrix is a trap.

Remark (Reading $A\mathbf{x} = \mathbf{b}$ through rank–nullity). A solution exists if and only if $\mathbf{b} \in \operatorname{im} T$. If $\mathbf{x}_p$ is one solution, every solution is $\mathbf{x}_p + \mathbf{z}$ with $\mathbf{z} \in \ker T$ (indeed, $A\mathbf{x} = \mathbf{b} = A\mathbf{x}_p$ gives $A(\mathbf{x} - \mathbf{x}_p) = 0$): the solution set is either empty or a shifted copy of the kernel. It is a single point exactly when the nullity is zero.

Remark (Common traps).

- The kernel lives in the domain, the image in the codomain. For $T\colon \mathbb{R}^3 \to \mathbb{R}^2$, $\ker T \subseteq \mathbb{R}^3$ while $\operatorname{im} T \subseteq \mathbb{R}^2$; writing them in the same space is a type error.
- "Injective $\iff$ surjective" needs finite dimensions and $\dim V = \dim W$. It fails in infinite dimensions: differentiation on all polynomials is surjective but kills constants, while multiplication by x is injective but never produces a nonzero constant. Rank–nullity is the finite-dimensional reason the two coincide.

In code terms, the kernel measures information loss: a nonzero kernel means distinct inputs collide, so the input cannot be reconstructed from the output. The rank counts the effective degrees of freedom in the output. Real neural networks interleave linear layers with nonlinearities precisely to escape these rigid limits.

The next step adds eigenvalues: for a map $T\colon V \to V$, the *eigenspace* for λ is $\ker(T - \lambda I)$, and rank–nullity will tell us how large it can be.

2 Exercises

Exercise 2.1. Decide which of the following maps are linear. Justify your answer from the definition.

(a) $T_1\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T_1(x, y) = (2x + y, x - 3y)$;

(b) $T_2\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T_2(x, y) = (x^2, y)$;

(c) $T_3\colon \mathbb{R}^2 \to \mathbb{R}$, $T_3(x, y) = x - 2y$.

Exercise 2.2. Let $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ be the linear map with

$$T\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\0\\2\end{pmatrix}, \qquad T\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}0\\1\\1\end{pmatrix}.$$

Compute $T\left(\begin{smallmatrix}x\\y\end{smallmatrix}\right)$ for a general $(x, y) \in \mathbb{R}^2$, and write down the matrix A with $T(\mathbf{x}) = A\mathbf{x}$.

Exercise 2.3. Let $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ be given by $T(\mathbf{x}) = A\mathbf{x}$ with

$$A = \begin{pmatrix}1 & 0 & 2 & 1\\ 0 & 1 & 1 & 2\\ 1 & 1 & 3 & 3\end{pmatrix}.$$

Find a basis for $\ker T$ and a basis for $\operatorname{im} T$, and verify that $\dim \ker T + \dim \operatorname{im} T = 4$.

Exercise 2.4. Let $T\colon V \to W$ be linear and injective. Prove that if $v_1, \dots, v_k \in V$ are linearly independent, then so are $T(v_1), \dots, T(v_k) \in W$.

Exercise 2.5. Let V and W be vector spaces with $\dim V = \dim W = n < \infty$, and let $T\colon V \to W$ be linear. Prove that T is injective if and only if T is surjective.

Exercise 2.6. A linear layer of a program maps input vectors $\mathbf{x} \in \mathbb{R}^n$ to output vectors $A\mathbf{x} \in \mathbb{R}^m$. Explain, in the language of this lesson, what a nonzero vector in $\ker T$ means for reconstructing the input from the output. Then explain what the rank of A tells you about the effective number of output coordinates, and why stacking two such linear layers (with no nonlinearity in between) can never produce more expressive outputs than a single linear layer.

3 Solutions

Solution to exercise 2.1. (a) Yes, the map is linear. Let $u, v \in \mathbb{R}^2$, and $a, b \in \mathbb{R}$. We need to prove:

$$T_1(au + bv) = aT_1(u) + bT_1(v)$$

Let's check:

$$(2(au_1 + bv_1) + (au_2 + bv_2), au_1 + bv_1 - 3(au_2 + bv_2)) = a(2u_1 + u_2, u_1 - 3u_2) + b(2v_1 + v_2, v_1 - 3v_2)$$
$$(2au_1 + 2bv_1 + au_2 + bv_2, au_1 + bv_1 - 3au_2 - 3bv_2) = (2au_1 + au_2, au_1 - 3au_2) + (2bv_1 + bv_2, bv_1 - 3bv_2)$$
$$(2au_1 + 2bv_1 + au_2 + bv_2, au_1 + bv_1 - 3au_2 - 3bv_2) = (2au_1 + 2bv_1 + au_2 + bv_2, au_1 + bv_1 - 3au_2 - 3bv_2)$$

Furthermore, by seeing what the image of the basis vectors $(0, 1)$ and $(1, 0)$ are, we can define a matrix a such that $T_1(x) = Ax$. A is defined like so:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}$$

(b) No it is not linear. A linear map would satisfy $T_2(ax) = aT_2(x)$, but a counter-example with $x = (1, 1)$ and $a = 2$ shows:

$$(2^2, 2) \neq (2 \cdot 1^2, 2 \cdot 1)$$
$$(4, 2) \neq (2, 2)$$

(c) Yes it is a linear, Similar to T_2, except with a smaller rank. Let's prove it from the definition of linearity:

$$T(au + bv) = aT_1(u) + bT_2(v)$$
$$au_1 + bv_1 - 2(au_2 + bv_2) = a(u_1 - 2u_2) + b(v_1 - 2v_2)$$
$$au_1 + bv_1 - 2(au_2 + bv_2) = au_1 - 2au_2) + bv_1 - 2bv_2)$$
$$au_1 + bv_1 - 2(au_2 + bv_2) = au_1 + bv_1 - 2au_2 - 2bv_2$$
$$au_1 + bv_1 - 2(au_2 + bv_2) = au_1 + bv_1 - 2(au_2 + bv_2)$$

□

Solution to exercise 2.2. Let's find out where T puts the basis vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ using linearity.

Setting $u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$T(\frac{1}{2}u + \frac{1}{2}v) = \frac{1}{2}T(u) + \frac{1}{2}T(v)$$
$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$
$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{3}{2} \end{pmatrix}$$

$$T(\frac{1}{2}u - \frac{1}{2}v) = \frac{1}{2}T(u) - \frac{1}{2}T(v)$$
$$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$
$$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

Now that we know where the basis vectors are in the image, we can write down a general formula for $T\begin{pmatrix}x\\y\end{pmatrix}$ and find the matrix A such that $T(x) = Ax$. The general form is:

$$T\begin{pmatrix}x\\y\end{pmatrix} = (\frac{1}{2}x + \frac{1}{2}y, \frac{1}{2}x - \frac{1}{2}y, \frac{3}{2}x - \frac{1}{2}y)$$

And the matrix A is:

$$A = \begin{pmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & -\frac{1}{2}\\ \frac{3}{2} & \frac{1}{2}\end{pmatrix}$$

□

Solution to exercise 2.3. Let's row-reduce A:

$$A = \begin{pmatrix}1 & 0 & 2 & 1\\ 0 & 1 & 1 & 2\\ 1 & 1 & 3 & 3\end{pmatrix}$$
$$R_3 = R_3 - R_1$$
$$A = \begin{pmatrix}1 & 0 & 2 & 1\\ 0 & 1 & 1 & 2\\ 0 & 1 & 1 & 2\end{pmatrix}$$
$$R_3 = R_3 - R_2$$
$$A = \begin{pmatrix}1 & 0 & 2 & 1\\ 0 & 1 & 1 & 2\\ 0 & 0 & 0 & 0\end{pmatrix}$$

The pivot columns are columns 1 and 2, and $Ax = 0$ gives:

$$x_1 + 2x_3 + x_4 = 0$$
$$x_2 + x_3 + 2x_4 = 0$$

The kernel has this basis (4 unknowns with 2 equations gives 2 degrees of freedom):

$$\ker A = \text{span}\left\{\begin{pmatrix}-1\\1\\1\\-1\end{pmatrix}, \begin{pmatrix}2\\10\\2\\-6\end{pmatrix}\right\}$$

And the image has this basis:

$$\text{im}\, A = \text{span}\left\{\begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}\right\}$$

Then $\dim\ker A = 2$ and $\dim\text{im}\, A = 2$ and $2 + 2 = 4$. □

Solution to exercise 2.4. Knowing that T is injective tells us that no 2 distinct vectors have the same image under T. Now let's use the linearity of T to construct a proof by contradiction. Suppose that $T(v_1), \ldots, T(v_k)$ aren't linearly independent, i.e. there are some $v_a, v_b, v_c \in V$ and $a, b \in \mathbb{R}$ such that

$$aT(v_a) + bT(v_b) = T(v_c)$$

This would imply by linearity of T that

$$T(av_a + bv_b) = T(v_c)$$

And, since T is injective, that would mean that

$$av_a + bv_b = v_c$$

But since $v_1, \ldots, v_k$ is linearly independent, there's a contradiction. Thus, $T(v_1), \ldots, T(v_k)$ is linearly independent. □

Solution to exercise 2.5. First, let's assume T is not surjective. This would mean that its image doesn't span the codomain, so $\operatorname{span} \operatorname{im} T \neq W$. This would imply that $\dim \operatorname{im} T < n$, so by rank-nullity, we have:

$$\dim \ker T > 0$$

If the null space has dimension > 0 this means there exist multiple distinct vectors $v_1, \ldots, v_k$ where $T(v_1) = \ldots = T(v_k) = 0$, which would mean T is not injective. Thus T has to be surjective for it to be injective. □

Solution to exercise 2.6. If $\dim \ker T > 0$ that would make T non-injective, as we just saw in the previous exercise. This means the input can't be reconstructed from the output. Also, $\operatorname{rank} A = \dim \operatorname{im} A$, so if $\operatorname{rank} A < m$ that means that the coordinates of Ax are not linearly independent. Stacking these linear layers doesn't change rank-nullity, and since neither the rank or the nullity can be negative, they can't be $> m$ either. This means that no matter how much linear layers you stack with no nonlinearity in between, you will never get more expressive ($\operatorname{rank} A > m$) outputs. □

4 Feedback

Feedback (exercise 2.1). Your conclusions for all three maps are correct. In part (a), the matrix is also correct: its columns are the images of e_1 and e_2. The linearity calculation starts with $T_1(au + bv)$ only implicitly, so label the two sides or introduce the expression you are expanding. The same issue occurs in part (c), where the first displayed line names T rather than T_3 and then writes T_1 and T_2 on the right. Keep the map names consistent. There are also unmatched right parentheses after $au_1 - 2au_2$ and $bv_1 - 2bv_2$; remove these rather than leaving them as algebraic noise.

Part (b) is a good counterexample. To make the calculation fully transparent, identify the left side as $T_2(2x)$ and the right side as $2T_2(x)$, rather than only showing the resulting coordinates. Part (c) should not be described as similar to T_2 "with a smaller rank": T_2 is nonlinear, so its rank has not been defined in this lesson, and the reason T_3 is linear is the coordinate calculation you give.

Feedback (exercise 2.2). Your recovery of $T(e_1)$ and $T(e_2)$ from the two given values is correct, and the general formula for $T(x, y)$ follows correctly from those images. There is one sign error in the matrix: you found $T(e_2) = (1/2, -1/2, 1/2)^T$, but its third entry becomes $+1/2$ in your matrix instead of $-1/2$. Thus the displayed matrix does not represent the formula immediately above it.

When you revise, use each recovered standard-basis image as a column and check your matrix against both of the originally supplied values. The phrase "where the basis vectors are in the image" is also imprecise: you have found their images, not located the basis vectors inside $\operatorname{im} T$.

Feedback (exercise 2.3). The row reduction, pivot columns, equations for the kernel, and image basis are correct. In particular, you correctly took the image basis from the original matrix. Your two kernel vectors do belong to $\ker T$, and they are linearly independent: for example, their third and fourth coordinates form the matrix

$$\begin{pmatrix} 1 & 2 \\ -1 & -6 \end{pmatrix},$$

whose determinant is nonzero. They also have the right number for a two-dimensional kernel.

The missing step is the derivation of these vectors from the free variables. Set $x_3 = s$ and $x_4 = t$, solve the two displayed equations for x_1 and x_2, and show that the resulting general vector is a linear combination of your two claimed vectors. Without that calculation, the sentence about two degrees of freedom explains the expected dimension, but does not prove that this particular pair spans the entire kernel. The notation $\ker A$ is acceptable for a matrix map, but $\ker T$ matches the exercise's notation.

Feedback (exercise 2.4). The conclusion is correct, and your use of linearity followed by injectivity contains the central idea. However, the proof does not yet handle the definition of linear independence for the whole list. A dependent list need not contain a vector v_c that is a combination of just two other vectors v_a and v_b. It might instead satisfy a relation involving three or more vectors, so your three-vector equation only treats a special case.

Start instead with an arbitrary relation $\sum_{i=1}^{k} c_i T(v_i) = 0$, where $c_1, \ldots, c_k \in \mathbb{R}$. Use linearity and injectivity to obtain a relation among $v_1, \ldots, v_k$, and then state what their independence forces about every c_i. You can prove the result directly; contradiction is not necessary.

Feedback (exercise 2.5). You have correctly proved that injectivity implies surjectivity in the equal finite-dimensional setting. Writing $\operatorname{span}(\operatorname{im} T) \neq W$ is redundant because $\operatorname{im} T$ is already a subspace; the direct statement is $\operatorname{im} T \neq W$. The step from $\dim \ker T > 0$ to noninjectivity needs the reason that a nontrivial kernel contains a nonzero vector v, while $T(v) = 0 = T(0)$. Listing distinct vectors $v_1, \ldots, v_k$ does not itself establish that there are at least two.

More importantly, this is an "if and only if" exercise, and the converse is missing. You must also assume that T is surjective and use $\dim \operatorname{im} T = n$ together with rank–nullity to determine the kernel. In both directions, state explicitly where the equal dimensions are used, and use the kernel criterion for injectivity from the lesson.

Feedback (exercise 2.6). Your first observation about a nonzero kernel vector is on the right track, but make the collision explicit: if $v \neq 0$ and $T(v) = 0$, then $T(x + v) = T(x)$ for every input x. Thus x and $x+v$ are distinct inputs with the same output, which explains why the output cannot determine the input uniquely. Merely saying that the input "can't be reconstructed" could be read as a claim about practical computation rather than this mathematical nonuniqueness.

The rank discussion needs correction. The scalar coordinates of Ax are always real numbers; saying that they are "not linearly independent" when $\operatorname{rank} A < m$ is not the relevant statement. Instead, describe the image as a subspace of $\mathbb{R}^m$ of dimension $\operatorname{rank} A$: at most that many independent output degrees of freedom can vary, even though the output has m scalar entries. Also, rank–nullity constrains rank by the number of input columns; it does not by itself justify your claim involving m.

The final composition argument is not correct as written. Nonnegative rank and nullity do not imply that neither can exceed m, and "more expressive ($\operatorname{rank} A > m$) outputs" is impossible for a map $\mathbb{R}^n \to \mathbb{R}^m$: the rank is always at most m because $\operatorname{im} A \subseteq \mathbb{R}^m$. The rank is also at most n, by rank–nullity. Replace this part with an argument about the composition itself: two matrices A and B compose to another single matrix, and the image of the composition is contained in the image of the outer map. Explain why this bounds, rather than increases, the available output degrees of freedom.